



Optimization of pXRF instrumentation conditions and multivariate modeling in soil fertility attributes determination

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ABSTRACT

The increasing global demand for agricultural products, food, and energy requires a sustainable and secure approach to agricultural production. Soil nutrition is vital for enhancing agricultural productivity, and understanding the spatial variability of fertility attributes is crucial in soil fertility prevention. Traditional soil analysis methods are time-consuming, costly, and generate waste. Spectral-analytical techniques, such as energy dispersive X-ray fluorescence (EDXRF) combined with multivariate analysis, offer a faster, cost-effective, and environmentally friendly alternative. This study aimed to optimize measurement conditions for predicting soil fertility attributes using a commercial routine from a portable EDXRF (pXRF) equipment and compare its performance with benchtop EDXRF using Partial Least Squares Regression (PLS) modeling. The data fusion from different measurement conditions using low- and mid-level fusion approaches was explored to evaluate the models' synergy. 364 soil samples at three different depths were used. The study area covers 12 ha of Red Latosol and Red Nitosol. Ten fertility attributes were evaluated: exchangeable macronutrients (Ca^{2+} , Mg^{2+} , and K^{+}), available phosphorus (P_M), pH, potential acidity ($\text{H} + \text{Al}$), soil organic carbon (SOC), cation exchange capacity (CEC), sum of exchange bases (SB) and base saturation percentage (BSP). The results indicated that the measurement of the three experimental conditions tested is unnecessary, and the data fusion does not significantly improve the model's performance. The use of 15 kV and 12.15 μA without filters provided the best performance results for all attributes simultaneously. By the established RPD ranges, the SOC, CEC, SB, and Ca^{2+} models were considered very good for quantitative prediction ($\text{RPD} > 2.0$), while the K^{+} and Mg^{2+} models were considered good for quantitative predictions ($1.8 < \text{RPD} < 2.0$). The P_M , pH, BSP, and $\text{H} + \text{Al}$ models provided fair predictions, which may be used for evaluation and correlation ($\text{RPD} < 1.8$). Furthermore, the pXRF optimized condition provided results equivalent to those modeled with benchtop EDXRF data. These results highlight the potential of pXRF as an economical, rapid, and efficient alternative approach to the conventional method, and may provide valuable insight for decision-making related to the use of fertilizers and other soil correction products.

1. Introduction

Global population continue growth has led to an increasing demand for agricultural products, food, and energy. Consequently, an increment in agricultural production to promote sustainability and ensure food security has become imperative [1].

Soil fertility plays a crucial role in improving agricultural production. It depends on several factors, including periodic analysis [2] and

comprehensive knowledge of the spatial variability of fertility attributes [3]. Periodic analysis is fundamental in the evaluation and monitoring of chemical and physical attributes of agronomic interest. Access to high-density information on the soil fertility attributes spatial variability is vital in the decision-making process to promote better nutrition conditions for plants. For example, varying agricultural inputs and using liming practices according to local changes in soil properties may be economically and productively beneficial.

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Among the soil fertility attributes analyzed are soil organic carbon (SOC), macronutrients (Ca, Mg, K, and P) in exchangeable form, and fertility indicators such as cation exchange capacity (CEC), the sum of exchangeable bases (SB), base saturation percentage (BSP), pH, and potential acidity ($H + Al$). Traditionally, the samples are manually collected and sent to laboratories where acidic or basic extractions are carried out, using techniques that, although accurate, require hard sample preparation, are destructive, time-consuming, and use reagents. Consequently, analyzing a large number of samples becomes time- and financially consuming, in addition to generating waste [4,5]. Thus, the development of faster and cheaper methodologies based on green chemistry has become a priority [4].

Spectral-analytical techniques have proven to be a successful alternative approach to soil properties evaluation [2,4,6–12]. These spectroscopy-based techniques do not directly quantify soil attributes, however, may be associated with multivariate calibration to build models capable of generalizing the information contained in the spectra to predict the soil parameters. Although the method is an indirect approximation approach, it is faster, less expensive, and environmentally friendly when compared to conventional analyses [8,13]. From this perspective, energy dispersive X-ray fluorescence (EDXRF) associated with multivariate analysis has been successfully employed in the analyses of several soil attributes, among which are those previously mentioned [14,4,15–22].

EDXRF is a non-destructive atomic emission technique based on photoelectric effect for the characteristic X-rays production after atomic deexcitation. It provides the advantageous possibility of multielement analysis, quickly and at low cost [23,24]. With an appropriate detector, the elements present in the sample are identified and quantified by evaluating the peak intensities in the spectra. The determination of elements lighter than Al is affected by the low fluorescent yield and, consequently, the EDXRF is adequate to quantify the total inorganic concentration of elements with $Z > 13$. On the other hand, most EDXRF equipment is operated at an energy range in which the photoelectric effect is not the only predominant interaction phenomenon [25–27]. The radiation scattering by electrons in the analyzed sample is also likely to occur. Elements with atomic number lower than 8 have higher cross-sections for Compton scattering. Thus, the Compton scattering may provide information about light elements and the general composition of complex organic samples [28].

In most published studies which use EDXRF associated with multivariate analysis to predict soil attributes, the data matrix used consists in a set of elemental concentration values obtained from a quantitative routine of the instrument. Specifically, the elementary concentrations are obtained using the X-ray characteristic peak intensities. These values from the raw spectra are derived through a deconvolution process combined with a computational fitting. After that, the conversion to concentration data is performed by a closed empirical package dedicated to soil analysis or by fundamental parameters method [18,22,14]. This dataset is easily dealt with and pre-processed since it only contains a small number of variables, which are the elements of interest used in the modeling (in general is <15 variables) [14]. However, the use of concentration in the modeling has some drawbacks, such as the uncertainties of the peak area associated with the deconvolution process and spectra fitting, especially when dealing with characteristic X-ray peaks that are prone to spectral interference [14,29]. The possible loss of important implicit information contained in the spectral profiles, mainly the scattering effects, which carry relevant information for modeling some of the fertility parameters [4,14,30], and the fact that in some cases the quantitative XRF routines may not be accurate due to the complexity of the matrix effect of the soil samples [4,18,14] are the problems. While it is well documented that proper sample preparation (drying, control the particle size, and use appropriate sample containers) may mitigate most of the matrix effects influence, the effect of organic matter (OM) on quantitative XRF measurements is still poorly understood [31]. OM provides nutrients that stimulate plant growth, and

consequently, soils are often improved by adding OM in the form of compost or other soil amendments, increasing the organic matter content to 20% or more [32,33]. OM fractions of this magnitude may affect pXRF measurements, especially when operated in the soil modes routines with internal calibrations developed using silica matrices without organic matter [34]. As the X-ray fluorescence signal emitted from the sample surface is a function of its composition, higher OM levels from light elements (such as C, O, and H), may cause a dilution effect, resulting in lower count rates for heavier elements. It occurs because the lighter elements (C, O, and H the main OM compounds) in the soil increase Rayleigh and Compton scattering at the lower energy range of the spectrum [35]. In this context, the elemental concentration data obtained by the traditional soil routine of the XRF instrument may be underestimated. Therefore, the use of raw spectral data is recommended. The raw spectrum provides the lowest possible error associated with pXRF measurements and, due to the aforementioned drawbacks, may provide better performance and interpretability in the modeling process compared to the concentration data set [14]. Notwithstanding, several operating conditions are possible to be applied in the measurement of an X-ray spectrum in a soil sample. To efficiently quantify the total concentration of light and heavy elements in the same sample, different measurement conditions are required. As an example: 15 kV in the X-ray tube for light elements (from Na to Sc) and 50 kV for heavy elements (Ti to U) [27]. It implies in a longer measurement time. Conversely, the use of portable EDXRF (pXRF) equipment in this methodology is advantageous because it allows for faster measurements and lower operating costs when compared to the benchtop ones.

There are several studies using pXRF with multivariate analysis to predict soil fertility attributes [18,36,16]. Most of them uses the soil mode quantitative routine from the instrument which consider the measurements conditions using values close to 15 kV, 30 kV and 50 kV with 30 s measurement in each one. In addition, few studies compare benchtop EDXRF *versus* pXRF performances. In this sense, this research was motivated by the study about the redundant information in the raw spectra from three different operating conditions in a portable EDXRF equipment, aiming to find an optimized measurement condition to create local models to predict soil attributes and compare them with the modeling at EDXRF benchtop data. For that, raw spectra from a commercial quantitative pXRF routine of soil samples were modeled with the Partial Least Squares Regression (PLS) to predict soil fertility attributes and the optimized experimental condition was evaluated.

Considering that data from different measurements conditions will be evaluated, data fusion approach are interesting tools to be applied. Data fusion seeks to exploit the synergism among disparate data to improve predictive accuracy [37]. The combination of data may be carried out basically at three levels: low, mid and high. Low- and mid-level were used. In the low-level fusion, the model is built with data from all sources concatenated in a single matrix. Thus, the concatenated model performance allows the synergy evaluation among the different data sources in the property of interest prediction. The mid-level fusion requires further steps. Firstly, it extracts the relevant features from each data source separately. After that, the concatenation process is performed, and the new dataset generated is used for modeling. There are different ways to extract the relevant information *e.g.*, through principal component analysis (PCA) scores [37]. Although still little explored with XRF spectral data, the Predictive Analysis of Common Dimensions (P-ComDim) is an alternative to mid-level fusion methods. In this approach, the common information shared by each spectral class is identified, enabling the interpretation of its relevance for the prediction attribute of interest [38].

In this perspective, the raw spectra data from the three different measurement conditions were fused at the low- and mid-level approaches before the modeling. The worse performance of the fused compared to the individual models indicates the lack of complementarity between the data, allowing to discard the requirement to measure with three experimental conditions together.

Facing these motivations, the hypotheses elaborated in this study were: i) The pXRF spectra data modeled with PLS may generate local models capable of predicting soil fertility attributes with good performance and confidence considering the context of the dataset evaluated; ii) Models with fused data should not generate superior performance results than models built with individual data sets, enabling the optimized use of a single experimental condition for modeling; iii) pXRF spectra may produce results equivalent to those constructed with benchtop EDXRF spectra. The optimized models' performance was compared with those reported by Dos Santos et al. [22,14], in which the same sample area was explored.

2. Methodology

2.1. Soil sampling and reference analyses

The study was carried out in an agricultural area at Ribeirão Vermelho basin in Cambé municipality, Paraná State, Brazil. This is a research site to study and monitor erosion, management practices, and soil fertility attributes. The area comprises the latitudes between 23°09'59.70" and 23°09'50.14" S and longitudes 51°14'42.27" and 51°14'56.87"W. The region is located in the north of Paraná State and covers 12 ha. The average slope is 0.09 m m⁻¹, with altitudes ranging from 544 m to 514 m.

The soils were classified as Red Latosol and Red Nitosol [39] or Rhodic Ferralsol and Rhodic Nitosol [40], respectively. These systems are highly weathered from basalt and have a very clay texture (760 g kg⁻¹ clay), whose predominance is kaolinite, iron and aluminum oxyhydroxides. No-tillage has been used for over 17 years with scarification carried out on average every five years. During this period soybeans were cultivated in the summer and corn in the winter. Oats were sometimes grown in winter.

Sampling was carried out on a regular 28 × 28 m grid with 6 additional points (outside the grid). The samples were manually collected at three depths (0–5, 5–10, and 10–20 cm), totaling 394. In the laboratory, the samples were macerated and sieved through a 2 mm mesh, roots and rocks were removed. Then, the samples were uniformly dried at 40 °C for 48 h to eliminate moisture as a variable in the conventional and EDXRF analyses.

The analyzed soil fertility attributes were: exchangeable macronutrients (Ca²⁺, Mg²⁺, and K⁺), available phosphorus (P_M), pH, potential acidity (H + Al), soil organic carbon (SOC), cation exchange capacity (CEC), sum of exchange bases (SB) and base saturation percentage (BSP). All conventional analyses were carried out in the IDR Soil Analysis Laboratory in Londrina, Paraná, Brazil. The methodology used for quantification is reported by Refs. [4,22,14]. The values of fertility attributes quantified by conventional methods were organized in vectors *y* and were used as dependent variables in the models.

2.2. pXRF analysis

Approximately five grams of soil in loose powder form were accommodated in polyethylene XRF cups (n°0.1530, Chemplex Industries Inc., USA) sealed with Mylar film for XRF scanning in the Tracer 5i (Bruker Inc.) handheld equipment. This amount of sample resulted in around 10 mm sample thickness, being considered an infinite sample for the X-ray energy range considered.

The instrument was operated in the GeoExploration mode. It is an empirical package which is factory calibrated using several standard soil samples covering geologically relevant ranges for 48 elements from Mg to U. The output of the Geoexploration mode is a table with the elemental concentration and their uncertainties. Measurements in this mode are carried out in three phases: (i) 30 kV and 13 µA with 25 µm of Ti and 300 µm of Al filters (designed in this study as 30 kV); (ii) 50 kV and 22.85 µA with the aforementioned filters plus a 75 µm of Cu filter (50 kV); (iii) 15 kV and 12.15 µA without filters (15 kV). The elements

with *Z* < 20 were quantified using 15 kV through a Compton peak ratio. The excitation was performed in air atmosphere using primary X-rays generated from a Rh tube for 30 s for each phase. A Silicon Drift Detector (SDD) was used, and the spectra were recorded in 2048 channels. The spectra were normalized by tube current and detector live time, resulting in spectra units of counts of photons per second per microampere (cps µA⁻¹).

All samples were measured three times in different sample portions, shaking the XRF cup before each measurement. A mean spectrum was calculated for each sample. The pXRF spectra generated by each phase (measurement condition) and the elemental concentrations were organized into individual matrices *X* and were considered as independent variables to build the PLS individual local models.

To verify the performance of the GeoExploration quantitative mode, CS-M2, PTXRF-IAEA04, PTXRF-IAEA09, PTXRF-IAEA13, IPT32 and IPT42 standard reference material (SRM) were evaluated. The concentration result for 20 chemical elements was reported. The average elemental concentration in the SRMs measured ten times were calculated and the average recovery, in %, were: Mg (243), Al (109), Si (93), P (141), S (133), K (97), Ca (95), Ti (110), Cr (155), Mn (70), Fe (109), Co (70), Ni (97), Cu (55), Zn (88), Ga (74), Rb (92), Sr (101), Y (95) and Zr (106). The elements with concentration recoveries between 80 and 120% were used in the modeling with concentration results.

2.3. Multivariate analysis methods

Six calibration strategies have been performed: (i) using the raw spectral data from 15 kV condition; (ii) using the raw spectral data from 30 kV condition; (iii) using the raw spectral data from 50 kV condition; (iv) using elemental concentration data reported by GeoExploration mode (CNC); (v) concatenation of 15 kV, 30 kV, and 50 kV raw spectra (CNT); and (vi) using scores of the P-ComDim analysis (PCD).

The complete dataset was divided, with the Kennard-Stone algorithm [41], into a calibration (70% of samples) and prediction (30% of samples) sets. The methodological steps followed in this study are shown in Fig. 1.

2.3.1. Individual models

PLS was used to build the prediction local models with individual spectra and concentration data. The PLS establishes a quantitative association between a matrix *X* of independent data (spectra matrix) and a vector *y* of dependent data (fertility attributes matrix) so that the maximum covariance between *X* and *y* is reached [42]. Thus, the PLS maintains a compromise between explaining the variance in *X* and obtaining the highest correlation with *y* [43]. To properly select the number of latent variables (LV) cross-validation using contiguous blocks with 10 splits was applied. The number of LV was chosen according to the minimum root mean square error of cross-validation (RMSECV). The identification and removal of outliers in all models was performed in agreement to the criteria of data with extreme leverage and unmodeled residuals [44]. All PLS modeling were performed in MATLAB R2022a (The MathWorks Inc., Natick, USA) and Rstudio software.

Two parts of all raw spectra were cut. The first corresponds to a region with a signal added by the equipment manufacturers while the second corresponds to a background region without characteristic peaks. As a result, 0.98 to 15.98 keV, 0.98 to 25.98 keV and 0.98 to 25.98 keV were the energy ranges used in the raw spectra of the 15, 30 and 50 kV instrument conditions, respectively. Thus, the 15 kV spectra had 750 variables while the 30 and 50 kV had 1250.

The 15 kV, 30 kV, and 50 kV raw spectra were pre-processed using Poisson scaling combined with Mean Center (Poisson + MC) to modeling. It was chosen after evaluation the best results reported by Dos Santos et al. [22]. Poisson pre-processing weight the data according to its particular uncertainty [45] and is justified because in the XRF data, the statistical uncertainty in each variable is governed by Poisson statistics [27], where the absolute uncertainty at any given data point is

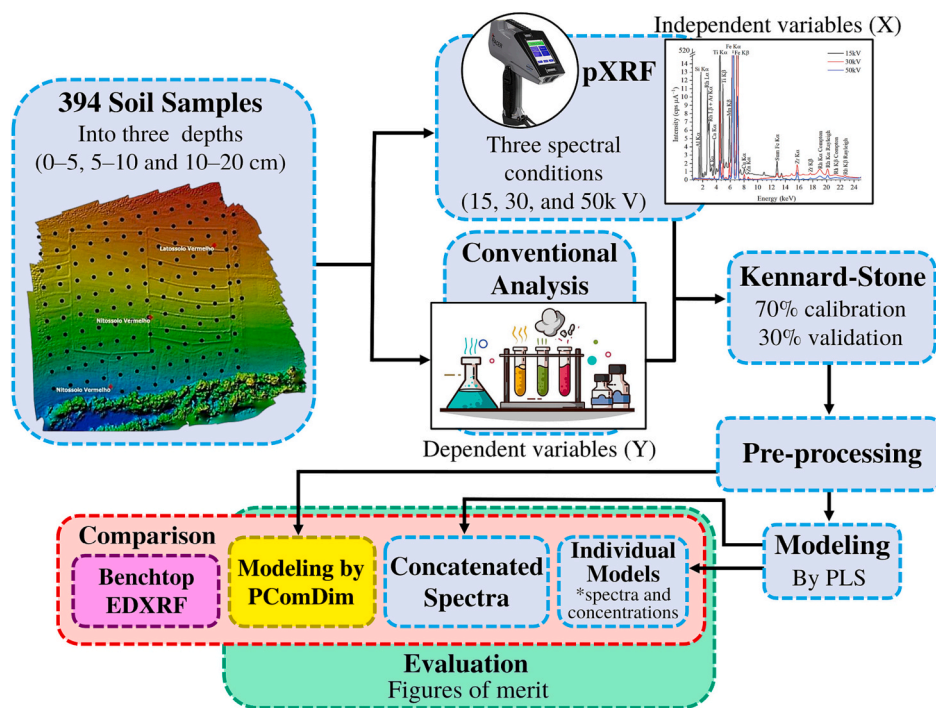


Fig. 1. Methodology schematic design.

larger as the number of counts represented by that point increases [46]. Thus, all the dataset variables are equalized, reducing the impact of variables that present great magnitude and mask information in the other variables with small magnitude.

According to the established criteria, the elements considered in the CNC model were Al, Si, K, Ca, Ti, Fe, Ni, Rb, Sr, Y, and Zr. To normalize the different orders of concentration data, autoscaling was the pre-processing applied.

2.3.2. Fusion models

Two data fusion approaches were used to build the fusion models. In the low-level approach, raw spectra were concatenated and PLS was performed (these models are called CNT). Before being concatenated, each spectrum was normalized by the Fe peak intensity to minimize differences among the characteristic peaks generated by the three configurations in the X-ray tube. The concatenation was carried out with the matrices arranged in the 15 kV, 30 kV and 50 kV order. Then, Poisson + MC pre-processing was applied.

The mid-level fusion was performed with the P-ComDim approach (these models are called PCD). The P-ComDim is a multi-block supervised method to explore the correlation between data from samples analyzed using different techniques (blocks) and a vector with a property of interest ($\mathbf{y}$) from the same samples. Latent variables, called “common dimensions” (CD), are derived using the singular value decomposition (SVD) method, highlighting the relationships between $\mathbf{y}$ and $\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_b$ ($\mathbf{X}_b$ represents the data from the b -th block). This “common singular value decomposition” (because it is related to $\mathbf{X}_b$ and $\mathbf{y}$) will involve not only a component common to the X-blocks ($\mathbf{t}$) but also a component in the $\mathbf{y}$ -space ($\mathbf{u}$). Then, these CDs are used in a regression model to predict $\mathbf{y}$ from $\mathbf{X}_1, \dots, \mathbf{X}_b$ [38,47].

Specifically, P-ComDim calculates the scalar product matrices $\mathbf{W}_b = \mathbf{X}_b \mathbf{X}_b^T \mathbf{y} \mathbf{y}^T$ to determine the common space among the X-blocks and $\mathbf{y}$, computing weights (called saliences) that express the importance of each block for each CD. This multi-block method is represented by [9,38,47,48]:

$$\mathbf{W}_b = \mathbf{Q} \mathbf{\Lambda}_b \mathbf{Q}^T = \sum_{i=1}^n \lambda_i^{(b)} \mathbf{q}_i \mathbf{q}_i^T \quad (1)$$

where $\mathbf{W}_b$ are the cross-product matrices corresponding respectively to the data blocks $\mathbf{X}_b$ and $\mathbf{y}$, which are first column-centered and then normalized (division by its Frobenius norm). $\mathbf{Q}$ is the matrix whose columns are the global score vectors $\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_i$ of common dimensions (CDs), and $\mathbf{\Lambda}_b$ is a diagonal matrix whose elements are denoted by $\lambda_i^{(b)}$, the salience of the b -th block for the i -th common dimension. The first common component for the X-blocks, i.e., $\mathbf{t}_1$, as well as a component in $\mathbf{y}$ -space, i.e., $\mathbf{u}_1$, are determined together with the specific saliences $\lambda_1^{(b)}$ by minimizing the criterion [9,38]:

$$\sum_{i=1}^b \| \mathbf{W}_i - \lambda_i^{(b)} \mathbf{q}_i \mathbf{q}_i^T \|^2 \quad (2)$$

Then, the next $(i + 1)$ -th CD parameters (such as saliences and scores) are calculated sequentially by the same algorithm, however, with a deflation of both X-blocks and $\mathbf{y}$ concerning $(i + 1)$ -th common component for the X-blocks ($\mathbf{t}_{i+1}$).

As aforementioned, for each common dimension extracted, there is a set of saliences, and scores. The scores are projections of the samples in the CD common space and, in the end, are used in a multiple linear regression to predict $\mathbf{y}$ [38,47]. For more details on this algorithm see Ref. [9,38,47].

The PCD was applied in the 15 kV, 30 kV, and 50 kV raw spectra pre-processed with Poisson + MC to verify the existence of common information shared by them. Moreover, it was also evaluated which spectral configuration most contributes to the prediction each fertility attribute. The number of common dimensions (CDs) was chosen according to the total accumulated variance and the best R^2 and root mean square error of prediction (RMSEP) values. The Graphical User Interface for P-ComDim analyzes used in this study was developed and coded for Matlab software by Mishra et al. [49] and may be downloaded at <https://github.com/puneetmishra2/Multi-block.git>.

Through the results generated by the data fusion approaches, the veracity of the second hypothesis may be attested.

2.4. Comparison parameters and figures of merit

The model's performance was evaluated and compared by root mean square error (RMSE), square correlation coefficients (R^2), and the ratio of performance to deviation (RPD) (Eq. 3).

$$RPD = \frac{\sqrt{\frac{\sum_{i=1}^n (\bar{y}_i - y_i)^2}{n}}}{RMSEP} \quad (3)$$

where y_i represents the i -th soil fertility attributes determined by conventional analyses, $\bar{y}_i$ the mean of the soil fertility attributes determined by conventional analyses and RMSEP the RMSE of prediction.

As reported by Viscarra Rossel et al. [50]: (1) excellent models (RPD > 2.5), (2) very good models (2.5 > RPD > 2.0), (3) good model (2.0 > RPD > 1.8), fair (1.8 > RPD > 1.4), and very poor model (RPD < 1.4).

The equivalence between the models was performed in two ways. At the first, the relative performance of the models was evaluated in percentage terms using the relative improvement (RI), calculated according to Eq. 4

$$RI = \frac{100(RMSEP_1 - RMSEP_2)}{RMSEP_1} \quad (4)$$

where RMSEP₁ is the RMSEP of the model taken as reference and RMSEP₂ is the value to be compared.

The statistical significance of the difference between the models' accuracy was evaluated by a randomization test with 0.05 significance [51,52]. This test compares the RMSEP values evaluating the null hypotheses in which the predictive accuracy is equal (RMSEP_A = RMSEP_B) and the alternative hypothesis in which the predictive accuracy is not equal (RMSEP_A ≠ RMSEP_B).

The experimental condition that produced the best individual models was identified according to performance comparison parameters. After that, the equivalence between models in the optimized experimental condition and fusion models was evaluated.

Moreover, the optimized condition models were also compared in terms of performance and equivalence with models built with spectral data from benchtop EDXRF, reported by Dos Santos et al. [22,14]. The benchtop PLS models were built with the same samples.

To attest the performance and quality of the optimized model, the following figures of merit were calculated: accuracy, systematic errors (bias), linearity, sensitivity (SEN), inverse of analytical sensitivity (γ^{-1}), limit of detection (LOD), and limit of quantification (LOQ) according to Ref. [44,51,53]

The accuracy was evaluated by RMSEP, RPD and relative error in the predicted values in percentage (RE%), Eq. 5

$$RE(\%) = 100 \sqrt{\frac{\sum_{i=1}^{n_p} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{n_p} y_i^2}} \quad (5)$$

where n_p is the number of prediction samples.

IUPAC defines bias as the difference between the population mean and the true value [54]. The presence of significant bias in the models was evaluated by applying a t -test as reported by Ref. [55]

Linearity was assessed using the Durbin-Watson (DW) test, according to Ref. [51,53]. This test compares successive residuals evaluating the hypotheses: there is no correlation between them (null hypothesis) and there is correlation between them (alternative hypothesis). In this case, the probability associated with the DW statistics (Eq. 6) was estimated by p -value. p -values > 0.05 indicates the absence of non-linearities at the 95% confidence level.

$$DW = \frac{\sum_{i=1}^{n_c-1} (e_{i+1} - e_i)^2}{\sum_{i=1}^{n_c-1} e_i^2} \quad (6)$$

e_i is the i th residual and n_c is the number of calibration samples.

The sensitivity, inverse of the analytical sensitivity, LOD and LOQ were quantified according to the references [55,56].

3. Results and discussion

3.1. Conventional analysis

Soil fertility attributes descriptive statistics quantified by conventional analyzes are presented in Table 1.

The results demonstrate that the Kennard-Stone algorithm separated the samples from the calibration and prediction set preserving the data representativeness. The K^+ and P_M values show high variability (CV > 60%) while pH and CEC are low (CV < 12%). The CV values for the other attributes are of medium variability in the study area (12% < CV < 60%) [57]. This range in the attributes is naturally caused by the fertility gradient on the Ferrasol profile, commonly observed in areas under no-tillage management where the highest fertility is observed on the surface layers, decreasing with increase of the depth [58]. Conversely, the inputs and fertilizers used may also alter soil fertility attributes. Thus, the high CV values for P_M and K^+ may be also related to the in-line application of fertilizers, as well as the adsorption and mobility of these elements [59]. Furthermore, the values for fertility attributes quantified in Ferrasols managed with a no-tillage system by Briedis [60] are similar to those presented.

3.2. pXRF results

The average pXRF spectrum from all soil samples is shown in Fig. 2. It is noted that the spectral response is different according to XRF instrumental setup. The 15 kV condition shows the highest background because none filter was used. Despite that, this condition is more efficient to excite elements lighter than Ca. However, there is not enough energy to excite elements heavier than Zn. At 30 kV and 50 kV conditions, the fluorescence of the elements heavier than Ti is favored and the Rh scattering K-lines peaks from the tube are evident. Furthermore, the background is lower under these two conditions due to the use of filters.

The descriptive statistics of the concentration results generated directly from Geoexploration routine are presented in Table 2. The concentrations of Fe, Al, and Si were quantified at higher levels. This may be attributed to soil characteristics, including the soil profile (rich in iron and aluminum oxyhydroxides), and high mineral fraction (the basis of the structure of most clay minerals is Si [61]). On the other hand, soils in an advanced state of weathering show intense loss of silica (composed mainly of SiO₂), with a resulting accumulation of insoluble iron and aluminum oxides [62], the most frequent being goethite (FeOOH), hematite (Fe₂O₃) and gibbsite (AlO₃H₃) [63]. In lower orders, the contents of Ti, Mn, Ca, and K were also quantified in larger quantities in relation to the remaining elements. In the analyzed soil, these may be related to several sources, such as the clay fraction mineralogy, soil texture, organic matter, and fertilizers.

3.3. Prediction models comparison

Fig. 3 shows the RMSEP, RPD, and pred R^2 results. It compares all the models for each fertility attribute. The prediction results were considered very good to poor depending on the fertility attribute (R^2 pred from 0.13 to 0.81 and RDP from 1.0 to 2.3). The models with highest pred R^2 and RDP values were SOC, CEC, SB, Ca²⁺, K⁺ and Mg²⁺. They also showed the lowest values for RMSEP.

Table 1
Descriptive statistics of soil attributes determined by conventional methods.

Attribute	Set of samples	Mean $\pm$ SD	CV (%)	Median	Minimum	Maximum
SOC (g kg ⁻¹)	All	19.8 $\pm$ 4.3	21.8	20.0	7.6	30.5
	Cal	19.8 $\pm$ 4.5	22.5	20.4	7.6	30.5
	Pred	19.9 $\pm$ 4.0	20.3	19.9	9.7	29.3
CEC (cmol _c kg ⁻¹)	All	13.7 $\pm$ 1.5	11.3	14.0	9.3	17.9
	Cal	13.8 $\pm$ 1.6	11.4	14.0	9.3	17.9
	Pred	13.5 $\pm$ 1.4	10.7	13.5	9.8	17.2
SB (cmol _c kg ⁻¹)	All	8.3 $\pm$ 1.9	22.7	8.0	2.1	12.6
	Cal	8.4 $\pm$ 1.9	22.2	8.7	2.2	12.6
	Pred	7.9 $\pm$ 1.9	23.6	8.2	2.1	11.5
Ca ²⁺ (cmol _c kg ⁻¹)	All	6.0 $\pm$ 1.5	25.2	6.0	1.4	9.2
	Cal	6.1 $\pm$ 1.5	24.5	6.4	1.4	9.2
	Pred	5.7 $\pm$ 1.5	26.4	5.7	1.4	8.9
K ⁺ (cmol _c kg ⁻¹)	All	0.24 $\pm$ 0.16	66.78	0.00	0.05	1.15
	Cal	0.3 $\pm$ 0.2	64.6	0.2	0.05	1.2
	Pred	0.20 $\pm$ 0.14	70.34	0.16	0.05	0.7
Mg ²⁺ (cmol _c kg ⁻¹)	All	2.0 $\pm$ 0.6	28.5	2.0	0.7	3.7
	Cal	2.0 $\pm$ 0.6	29.0	2.0	0.7	3.7
	Pred	2.0 $\pm$ 0.6	27.2	2.0	0.7	3.5
P _M (mg kg ⁻¹)	All	14.0 $\pm$ 10.0	71.6	12.0	0.4	69.8
	Cal	13.4 $\pm$ 8.9	66.7	12.3	0.4	60.3
	Pred	15.3 $\pm$ 12.1	78.8	12.0	1.3	69.8
pH	All	5.3 $\pm$ 0.4	7.7	5.0	4.2	6.5
	Cal	5.3 $\pm$ 0.4	7.6	5.2	4.2	6.5
	Pred	5.2 $\pm$ 0.4	8.0	5.2	4.2	6.5
BSP (%)	All	60 $\pm$ 10	16.1	61.0	18.0	80.0
	Cal	60.2 $\pm$ 9.3	15.4	61.5	19.9	79.7
	Pred	58.1 $\pm$ 10.2	17.6	60.4	17.7	77.2
H + Al (cmol _c kg ⁻¹)	All	5.5 $\pm$ 1.1	19.7	5.0	2.7	9.7
	Cal	5.4 $\pm$ 1.1	19.7	5.3	2.7	9.0
	Pred	5.6 $\pm$ 1.1	19.9	5.6	2.9	9.7

Ca, K, Mg, and P concentration values represent the bio-disponible fraction. SD = Standard deviation; CV = coefficient of variation; All = complete dataset with 394 samples; Cal = calibration set; Pred = prediction set.

Considering the individual raw spectra modeling, the 15 kV PLS models were the best for predicting SOC, CEC, and K⁺. For SB, the best model was obtained with 30 kV while for Ca²⁺ was with 50 kV. However, in both situations, the 15 kV spectra produced equivalent results. All the Mg²⁺ spectral models showed similar results, with the concentrations models being the best. The number of samples, the number of latent variables (LV), RMSE, and R² values of calibration, cross-validation, and prediction for all individual PLS models are available

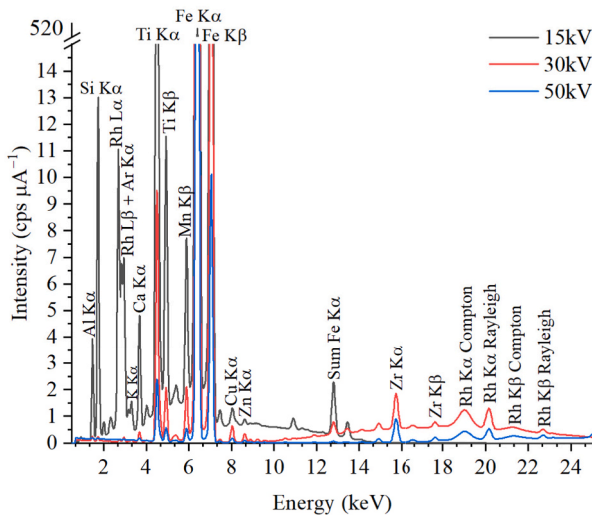


Fig. 2. Mean spectra of the 394 soil samples using three experimental conditions in the pXRF spectrometer.

at Table S1 in the supplemental material.

According to the RPD intervals established by Viscarra Rossel et al. [50], the 15 kV SOC, 15 kV CEC, 30 kV SB, 50 kV Ca²⁺ and CNC Mg²⁺ models were considered very good for prediction (RPD greater or equal to 2.0) while the 15 kV K⁺ model presented RPD of 1.9, considered good for predictions. Quantitative analyses are reliable for these attributes. These findings support the first hypothesis, indicating that the pXRF raw spectra data modeled with PLS may generate local models capable of predicting these fertility attributes analyzed in the context of this study with good performance and confidence.

In the remaining attributes, the 15 kV model produced the best results for P_M while CNT, CNC and 50 kV was the best for pH, BSP and H + Al, respectively. These four spectral models showed lower pred R² and RPD results when compared to the previous. The RPD values ranged from 1.7 and 1.4, indicating fair predictions. The P_M, pH, BSP, and H + Al results may be used in the qualitative evaluation and correlation analysis. These soil attributes depend on many factors, including soil texture, mineralogy, organic matter, soil depth, and water availability. The magnitudes variation of these parameters in the soil is not accurately captured by the pXRF spectrum. In the pH and BSP cases, the lowest CV values in the conventional analysis (Table 1) attest to the low variability of these attributes captured by the samples, making their prediction even more difficult by the adopted approaches. Conversely, although total P signal is detected by XRF, the modeling of P_M is challenging for several reasons. One reason may be associated with the low photoelectric cross-section and P fluorescent yield, as well as its low concentration in the soil compared to other elements such as Si and Al. Another reason may be the interference of other elements in P quantification. Phosphorus emits characteristic X-rays at two main energies: 2.013 keV (Kα) and 2.139 keV (Kβ). More abundant elements in soil, such as Al and Si, may absorb these characteristic X-rays before reaching the detector, since they have absorption edges close to these energies (1.559 and 1.839 keV for Al and Si respectively). Other studies with similar methods also have shown difficulty in predicting P_M, even with XRF and NIR data [3] and presented close results for other fertility attributes [4,10,22,14].

Regarding the CNC models, the better were SB, P_M, pH, BSP, and H + Al. However, the concentration data models presented equal or inferior performance than the individual spectral models, with higher RMSEP and lower R² prediction in most soil attributes, except for Mg²⁺ and BSP models. The presence of different characteristics in the samples (such as different soils, and other organic and mineral materials) affects the total elemental concentration measurements, but the spectral data are more

Table 2

Descriptive statistics of the total concentration per element determined by pXRF.

Element	Mean $\pm$ SD	Med	CV	Max	Min	Q25	Q75	Kurt	Skew
Si (g kg ⁻¹)	98 $\pm$ 10	99	10%	127	67	91	104	0.04	-0.3
Al (g kg ⁻¹)	100 $\pm$ 8	100	8%	126	76	95	105	0.1	-0.1
K (g kg ⁻¹)	0.68 $\pm$ 0.13	0.67	19%	1.23	0.39	0.59	0.78	0.08	0.44
Ca (g kg ⁻¹)	1.7 $\pm$ 0.6	1.6	35%	4.4	0.6	1.3	2.0	2.2	1.2
Ti (g kg ⁻¹)	22 $\pm$ 3	22	14%	33	15	20	24	1.0	0.5
Mn (g kg ⁻¹)	1.6 $\pm$ 0.3	1.5	19%	2.4	0.9	1.3	1.8	-0.5	0.5
Fe (g kg ⁻¹)	188 $\pm$ 13	190	7%	219	151	182	196	0.3	-0.6
Zn (mg kg ⁻¹)	94 $\pm$ 11	93	12%	144	71	87	101	0.7	0.7
Rb (mg kg ⁻¹)	14 $\pm$ 4	13	29%	24	8	11	18	-0.8	0.6
Sr (mg kg ⁻¹)	12 $\pm$ 3	12	25%	35	6	10	14	5.8	1.5
Y (mg kg ⁻¹)	42 $\pm$ 7	41	17%	68	23	36	47	0	0.2
Zr (mg kg ⁻¹)	226 $\pm$ 13	226	6%	259	181	217	235	0.2	-0.3

SD = Standard deviation, Med = median, CV = coefficient of variation, Var = Variance, Max = Maximum, Min = Minimum, Q25 = First quartile, Q75 = Third quartile, Kurt = Kurtosis, Skew = Skewness.

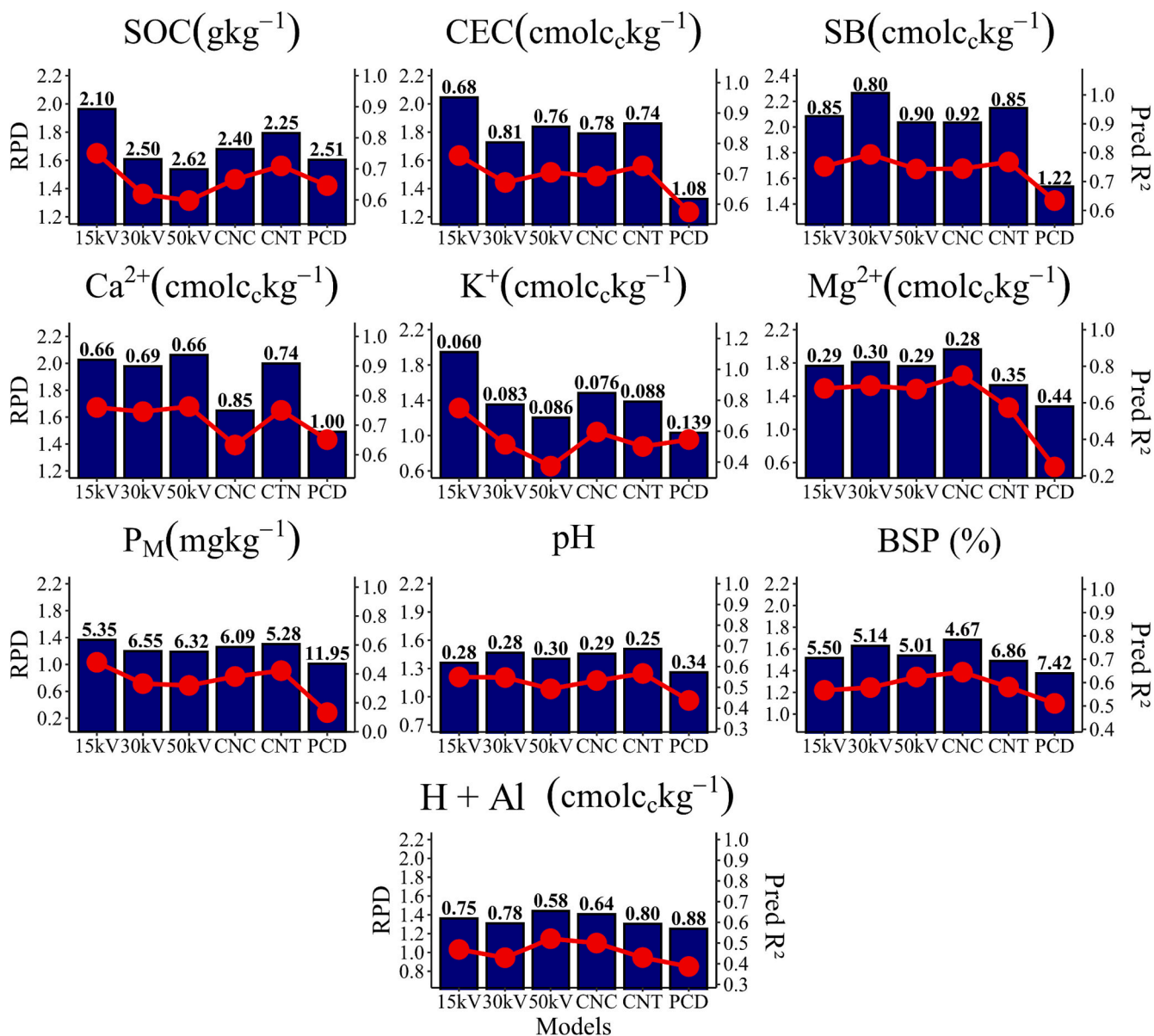


Fig. 3. RPD (blue bars and right vertical scale), prediction R^2 (red circles and left vertical scale), and RMSEP (bold values at top of bars) for all prediction models. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

sensitive to these variations. This complete representation of the complex matrix existing in a soil sample provided by spectral data may be a possible reason to the quantitative data models inferior performance compared to the spectral ones.

Concerning fusion models, the use of concatenated spectra resulted in predictions similar or lower than models with individual spectra attested by the RPD values ranging from 1.3 to 2.1. Table S2 in the supplementary material presents the complete details. The PCD models' saliences and regression coefficients values of each CD for each soil attribute are presented in Fig. 4. It is possible to verify that all spectral conditions have similar salience values in each common dimension for

all soil attributes, which indicates that the common information generated by the different conditions is distributed across the CDs. Therefore, redundant information is contained in the spectra obtained at 15, 30, and 50 kV. Consequently, there are no regression coefficients with values much higher than the others in each CD. Reflecting this spectra information redundancy, the PCD models are also inferior to the individual models. The RPD varied between 1.0 and 1.6 (Fig. 3), with better results for the SOC, SB, and Ca^{2+} models, classified as fair. The other attributes were below 1.4, being considered very poor models. The complete details of PCD models are at Table S3 in the supplementary material.

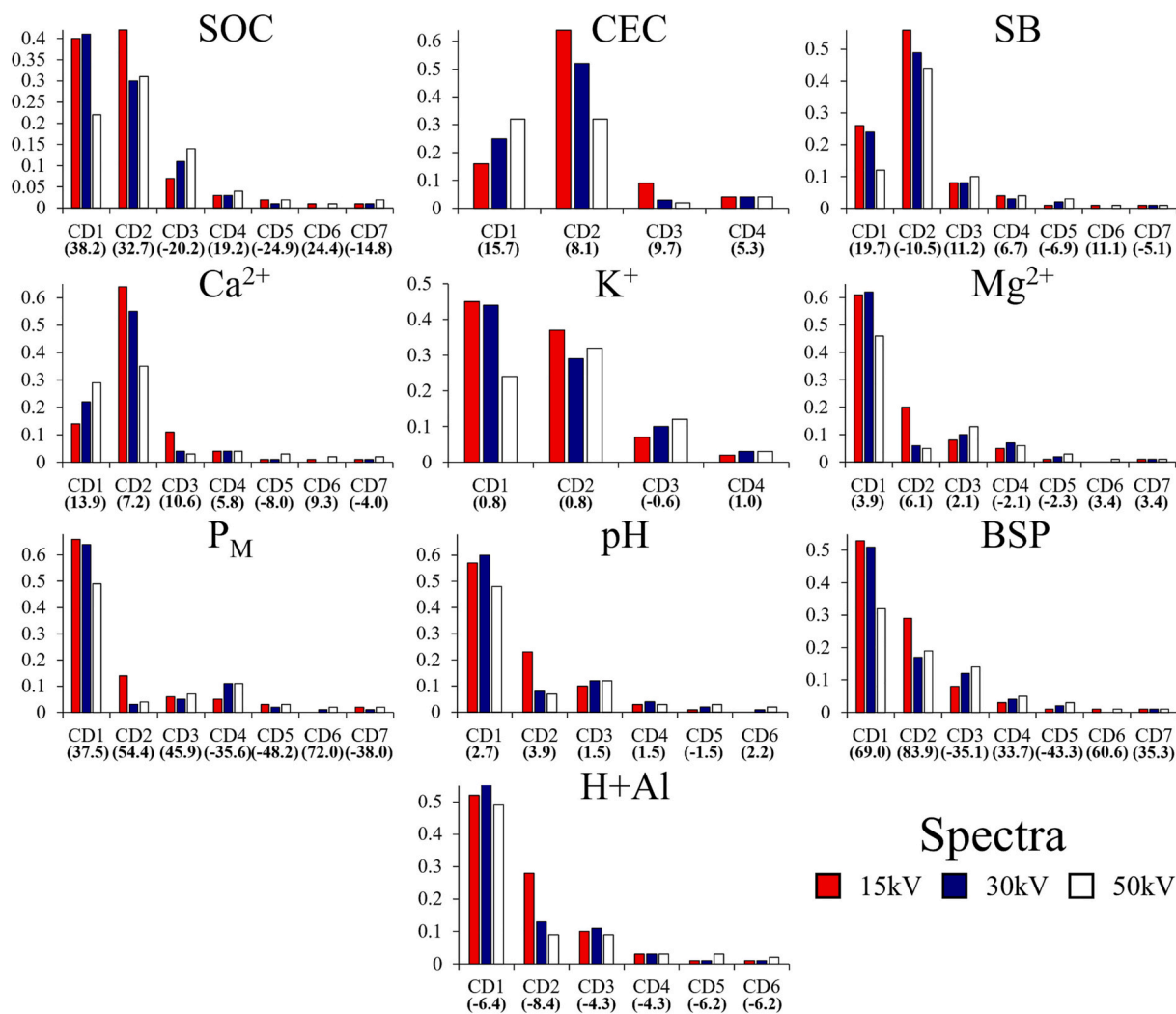


Fig. 4. Saliences (vertical bars) and regression coefficient values (below the CDs on the X axis) of the P-ComDim models.

Table 3

Random Test p-values comparing individual PLS 15 kV models with all other models for each fertility attribute.

Attribute	15 kV × 30 kV	15 kV × 50 kV	15 kV × CNC	15 kV × CTN	15 kV × PCD
SOC	0.004	0.001	0.01	0.03	0.001
CEC	0.03	0.01	0.03	0.2	0.001
SB	0.3	0.2	0.2	0.4	0.001
Ca^{2+}	0.2	0.4	0.002	0.1	0.001
K^{+}	0.001	0.001	0.01	0.01	0.01
Mg^{2+}	0.3	0.4	0.4	0.02	0.001
P_M	0.06	0.06	0.1	0.4	0.001
pH	0.03	0.2	0.9	0.9	0.07
BSP	0.2	0.2	0.07	0.04	0.01
H + Al	0.01	0.3	0.06	0.2	0.04

Bold indicates values greater than critical (0.05 for 95% confidence).

Evaluating all the model's performance in Fig. 3 it is verified that the 15 kV raw spectra produced the best models for a greater number of fertility attributes than the other experimental conditions. Therefore, 15 kV condition may be a general optimized condition. To attest it, the equivalence between the 15 kV model and all other models were performed with randomization tests. Table 3 presents the random test *p*-values.

The randomization test at 95% of confidence level demonstrates that 15 kV condition is considered equivalent to the best ones in the SB, Ca²⁺, Mg²⁺, pH, BSP and H + Al models. This may be justified because, unlike other spectral configurations, the use of 15 kV without filters in the X-ray tube is advantageous for soil analysis, as it favors the fluorescence of light elements [27], which are correlated with most of the analyzed soil fertility attributes. Moreover, in this configuration the Rh scattering L-lines from the X-ray tube target are also present. The scattering peaks provide subtle but significant information about the sample complex organic composition [28] and contribute to model the fertility attributes, once they are directly or indirectly influenced by light elements such as N, C and O in the soil organic composition. Thus, the 15 kV models were considered the optimized condition to be used as a reference in the pXRF spectrometer to predict all the fertility attributes for comparisons in the next subsections.

The RI values for the CNT models compared to the 15 kV model were: SOC (−7.0); CEC (−8.2); SB (0.4); Ca²⁺ (−12.7); K⁺ (−47.0); Mg²⁺ (−19.1); P_M (1.2); pH (−16.2); BSP (−24.7); H + Al (−6.7). Therefore, the concatenated spectra employment was only beneficial for the P_M modeling compared to the individual 15 kV model, with a 1.2% improvement. However, it was not reflected in a higher RPD value, not being enough to classify the model as good and suitable for quantitative analysis [50]. Similar results were found in previous studies using different spectroscopic sensors [3].

The PCD models compared to the 15 kV model only produced negative RI values: −22.34, −59.47, −42.89, −52.89, −118.46, −51.7, −123.28, −6.62, −34.89, −18.7 for SOC, CEC, SB, Ca²⁺, K⁺, Mg²⁺, P_M, pH, BSP, and H + Al, respectively. These negative values indicate that there was no relative improvement with this multiblock modeling. Despite the unsatisfactory results of the regression with P-ComDim, the methodology allowed verifying the existence of redundancy among the raw spectral data of the different measurement conditions in quantitative values terms.

The worse results of the data fusion models compared to the individuals allow discarding the need to measure under three experimental conditions. Only one experimental condition, in this case 15 kV, 12.15 μA, 30 s, without filters in the X-ray tube produced reliable prediction results for all soil fertility attributes evaluated simultaneously. These findings are evidence that directly corroborated the second hypothesis. The optimized use of a single experimental condition may produce a shorter time interval for measurements, a considerably smaller amount of data for manipulation and less processing power to manage the models.

It is important to mention that these negative data fusion results are related to the particularities of the dataset, which explores the data fusion from the same analytical technique, just with different experimental conditions. Data fusion seeks to exploit the synergy between different data to improve the predictive accuracy of models [37]. As a result: Studies from techniques based on different bands of the electromagnetic spectrum and with different objectives have shown that data fusion may improve predictions of various soil attributes [6,8,9]. Dos Santos et al. [9] fused EDXRF and visible and near-infrared (vis-NIR) data from soil samples from the same region evaluated in this study. With three fusion approaches (at low, medium, and high levels), the results showed an improvement in the modeling of SOC, Ca²⁺, Mg²⁺, pH, SB, and CEC compared to the best individual models. Tavares et al. [8] evaluated two data fusion approaches (at low and high levels) with samples from two agricultural fields with soils characterized as Lixisol and Ferralsol in modeling fertility attributes. The results showed that the

data fusion with visible and near infrared (VNIR) diffuse reflectance, XRF, and laser-induced breakdown spectroscopy (LIBS) may increase the number of fertility attributes predicted with optimal performance, as well as improve the prediction accuracy of Clay, OM, CEC, pH, BSP, and P extractable by exploiting the synergy of these data fusion techniques. Therefore, data fusion has shown greater potential to increase modeling accuracy when highly synergistic data are combined. However, while data fusion approaches are promising, they may also be costly from a mathematical and computational point of view. To avoid this, another approach that has shown promising results in improving the performance of fertility attribute modeling is auxiliary variables (AV). Sharma et al. [64] achieved positive results by modeling temperate soil CEC from pXRF data with Clay, pH, and organic matter as AV using a multiple linear regression model. Andrade et al. [16] showed that the inclusion of soil texture as AV with pXRF data may improve the prediction of exchangeable Ca²⁺, Mg²⁺, and available K. Dasgupta et al. [65] showed that the combination of agro-climatic, physiographic, and routine soil test parameters (pH, electrical conductivity, loss on ignition organic matter, and SOC) as AV with pXRF produced the best prediction accuracy for most soil fertility attributes among all tested (available N, −K, −Ca, −Mg, −Zn, −Fe, −Cu, −Mn, −B, K/Mg ratio, S availability index, and total exchangeable base). Nevertheless, although AV methodologies are not as mathematically and computationally costly compared to data fusion, the complexity involved in the process of obtaining these auxiliary parameters is significant, as many of them are extracted through conventional analysis, e.g., soil texture, pH, organic matter, among others. AV represents a current and interesting topic that requires a well-defined objective to be applied because, although studies have shown potential for increasing the predictive capacity of model performance, they may not be advantageous in optimizing methodology, reducing the number of samples sent for conventional analysis and from a financial point of view.

Conversely, the range of soil attribute values is an important issue, which directly impacts on the performance results. The models are local, so they are valid for the range of fertility attribute values used for modeling (Table 1). Consequently, the range of values may limit the effectiveness of the methodology for quantifying the soil attributes, especially those that are least related to the XRF data, e.g., pH. Further studies with greater variability in soil attributes is needed to effectively assess the performance limit of the method with XRF data.

Table 4

Figures of merit of the 15 kV PLS models to each soil attribute.

Attribute	p-value DW Test	Bias (t _{bias})	SEN	γ ^{−1}	RE (%)	LOQ	LOD
SOC (g kg ^{−1})	0.82	0.37 (1.95)	0.3	0.2	10	1.83	0.60
CEC (cmol _c kg ^{−1})	0.15	0.03 (0.50)	0.6	0.1	5	0.99	0.33
SB (cmol _c kg ^{−1})	0.5	0.07 (0.85)	0.7	0.1	11	0.89	0.29
Ca ²⁺ (cmol _c kg ^{−1})	0.5	0.05 (0.65)	0.4	0.08	11	0.80	0.27
K ⁺ (cmol _c kg ^{−1})	0.94	0.01 (2.46)	0.15	0.04	28	0.40	0.13
Mg ²⁺ (cmol _c kg ^{−1})	0.98	0.03 (0.93)	1.63	0.04	14	0.38	0.13
P _M (mg kg ^{−1})	0.85	0.06 (0.02)	0.22	0.14	36	1.44	0.47
pH	0.12	0.03 (1.28)	1.9	0.03	6	0.33	0.11
BSP (%)	0.41	0.50 (0.94)	0.66	0.09	9	0.94	0.31
H + Al (cmol _c kg ^{−1})	0.31	0.10 (1.15)	0.24	0.26	13	2.56	0.84

p_{critic} = 0.05; t_{critic} = 1.98.

3.4. Figures of merit and VIP score

Once established the optimized reference models, the figures of merit for the 15 kV models were calculated (Table 4). Through the values of $|Bias|$ and t_{Bias} it was observed that only the K^+ model showed significant bias ($t_{Bias} > t_{critic}$). The LOD and LOQ values were above the lowest values obtained by conventional analysis only for the K^+ and P_M models (Table 1). For the other fertility attributes, the entire range of values obtained by the reference method is valid.

Besides the RMSE and RPD values, RE(%) was also evaluated to certify the models accuracy. The K^+ and P_M models showed the highest RE(%), 28 and 34%, respectively. The other models presented values below 14%.

On the other hand, the scatterplots for the 15 kV models of all soil attributes (Fig. 5) show a linear pattern between predicted and reference data. Although pH, BSP and H + Al models show slightly different behavior than the linear fit line, the visual analyzes are qualitative ways of analyzing linearity, which was verified quantitatively through the DW test. The DW p -values at 95% confidence level did not indicate a lack of linearity in anyone model (Table 4). Thus, linear multivariate calibration methods such as PLS is enough for modeling soil fertility attributes with pXRF spectral data.

The spectra energy range or peaks with higher importance in the prediction of each fertility attribute for the 15 kV models were evaluated by the informative vector called variable influence on projection (VIP) scores. According to the VIP scores (Fig. 6), the Ca- α characteristic X-ray peak was the most important variable in SOC, SB, Ca^{2+} , Mg^{2+} , P_M , pH, BSP and H + Al models. The CEC and K^+ models had Mn- α , and K- α peaks, respectively, as the most important variables. However, the peak Ca- α is the second most important variable in these models. These

findings agree with the correlation analysis carried out in previous studies by Dos Santos et al. [14]. This was expected because Ca is directly or indirectly involved with most of the evaluated attributes. The pH variation over time happens according to soil management, successive crops and fertilizations. When plants absorb positively charged nutrients (including Ca^{2+}), they release H^+ from the roots into the soil solution, which reduces the pH. The Ca^{2+} and Mg^{2+} contents are directly related to soil acidity. Generally, acidic soils have low Ca^{2+} and Mg^{2+} levels and soils with good fertility have higher Ca^{2+} and Mg^{2+} levels [66]. Moreover, Ca^{2+} is directly considered in CEC, SB, and BSP calculations.

The significant importance of Ca content in different fertility attributes modeling has also been reported in other studies conducted in different Brazilian tropical soils [16,14,17,21,67,68]. Because a significant part of the mineral composition is quartz, micas, kaolinite and Fe and Al-oxides, Brazilian tropical soils may lack Ca in their crystalline structure. According to most authors, this may explain the Ca content in soils, since 95% of its speciation consists of the exchangeable form Ca^{2+} [14].

3.5. Comparison between pXRF and benchtop EDXRF models

The prediction results of the EDXRF Benchtop models reported by Dos Santos et al. [22,14] are shown in Table 5. Comparing the results of the 15 kV models with them, it is possible to verify that the values of Pred R^2 , RMSEP and RPD were similar, with a slight superiority for the pXRF models in the case of CEC, K^+ , Mg^{2+} , and P_M .

A randomization test at 95% confidence level comparing the 15 kV models' prediction values with the benchtop EDXRF values was performed. It is notable that, among all soil attributes analyzed, only the CEC and Mg^{2+} pXRF models were not considered statistically equivalent to the models built with benchtop EDXRF spectra. In these cases, the pred R^2 , RMSEP, and RDP values indicated a small superiority of the pXRF models over the benchtop EDXRF models [14]. Although the difference in the model's performance is small, it may be responsible for the non-equivalence in the randomization test.

In general terms, the equivalence between pXRF and benchtop EDXRF are evidence that corroborate the third hypothesis, indicating that PLS modeling with spectra from portable XRF equipment may generate local models as accurately and with good performance as the methodology with benchtop equipment. Since the benchtop equipment remains in the laboratory, a great advantage of using pXRF is the mobility associated with the measurement step, which provides a potentially possibility for *in situ* measurements. However, *in situ* measurements are complicated because they omit the essential sample preparation stage. Consequently, issues regarding soil moisture, granulometry, the presence of materials like gravel, other contaminants, and soil flatness irregularities, which are minimized by proper preparation, are evident [18,14,31]. These effects, combined with spectral interferences and OM attenuation that cannot be removed in advance, may significantly affect modeling performance. For all intents and purposes, the application of the *in situ* methodology is promising and still needs to be systematically evaluated.

Overall, the results of this study contribute to the development and establishment of a faster and cheaper method for quantifying fertility attributes on a local scale. In this context, after a robust calibration and validation of the models at the study area until they reach a pre-determined accuracy, measurements of soil attributes may be obtained only with the spectral reading of new soil samples [3,69]. On the other hand, as soil chemical mapping needs to be renewed periodically due to the influence of agronomic practices and the dynamics of the soil-plant-atmosphere system [70], this methodology may allow the soil fertility spatial variability to be quickly determined and with a smaller number of samples sent for conventional analysis. The decrease in sampling density based only on conventional laboratory analysis is advantageous from a financial, environmental, and logistical point of view. Thus, with

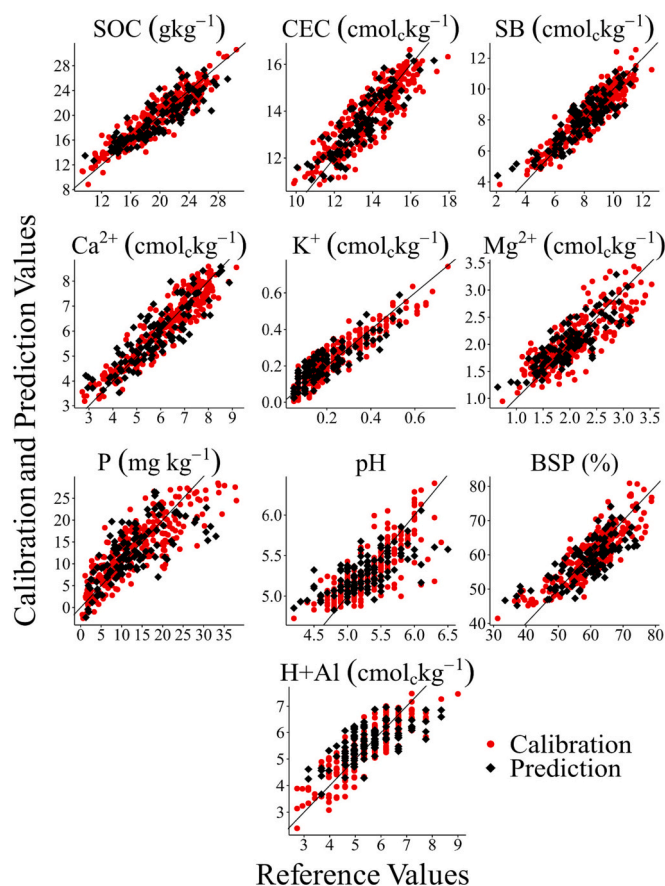


Fig. 5. Scatter plots of reference values versus predicted values for PLS 15 kV models.

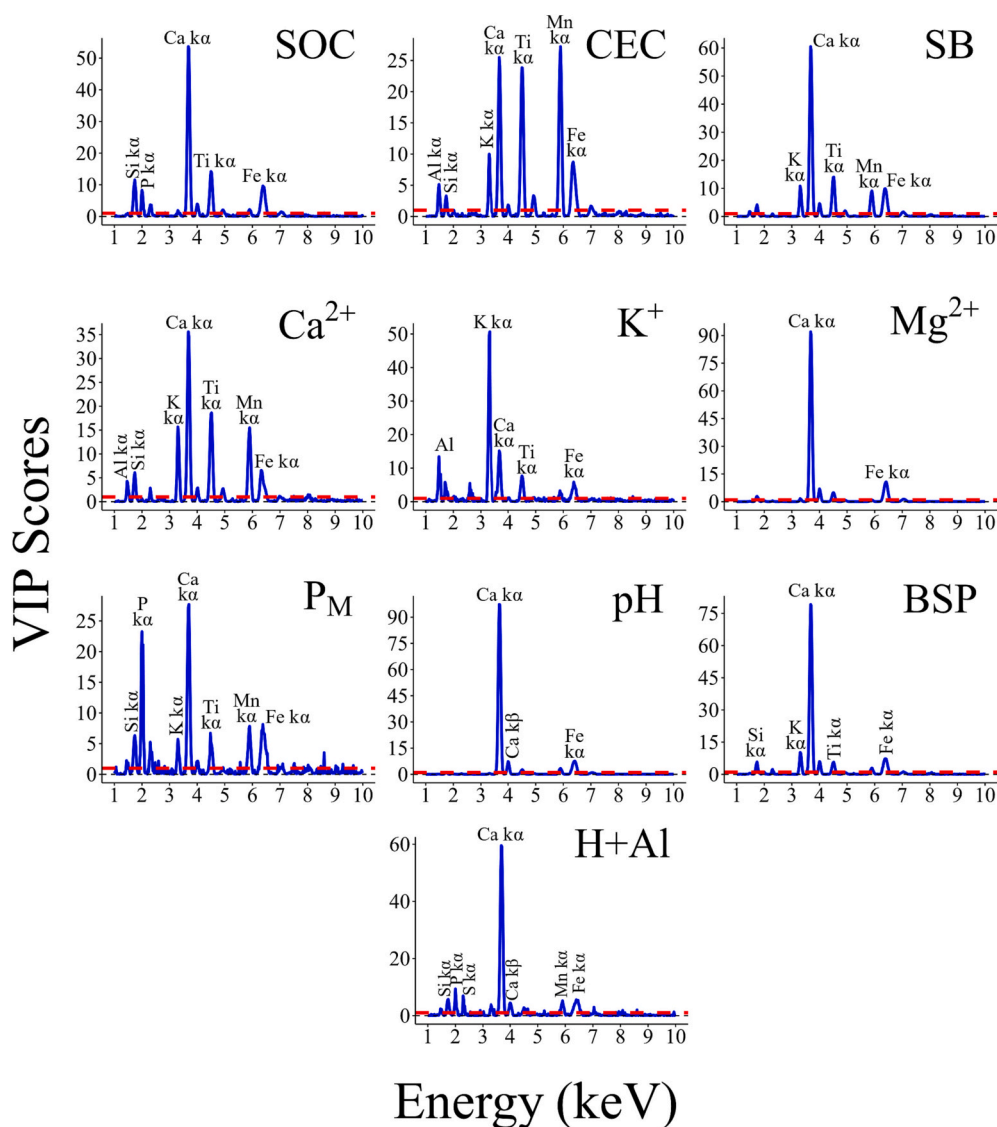


Fig. 6. Variable influence on projection (VIP) scores of the 15 kV PLS models to all fertility attributes. Values greater than unity (red dotted line) indicate variables that significantly contributed to modeling. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 5

Prediction results from PLS calibration models with benchtop EDXRF spectral data reported by Dos Santos et al. in [22,14].

Attribute	Pred R ²	RMSEP	RPD	Random Test p-value
SOC (g kg ⁻¹)	0.79	1.90	2.2	0.31
CEC (cmol _c kg ⁻¹)	0.67	0.85	1.7	0.01
SB (cmol _c kg ⁻¹)	0.82	0.77	2.4	0.15
Ca ²⁺ (cmol _c kg ⁻¹)	0.81	0.63	2.3	0.32
K ⁺ (cmol _c kg ⁻¹)	0.65	0.075	1.7	0.93
Mg ²⁺ (cmol _c kg ⁻¹)	0.64	0.35	1.7	0.02
P _M (mg kg ⁻¹)	0.39	5.72	1.2	0.19
pH	0.57	0.26	1.5	0.16
BSP (%)	0.71	5.2	1.8	0.45
H + Al (cmol _c kg ⁻¹)	0.55	0.72	1.5	0.27

Pareto scaling pre-processing was used, with iSPA variable selection in the SOC models and full spectrum for the others. The randomization test regarding the 15 kV pXRF model is presented. Bold indicates values greater than critical (0.05 for 95% confidence).

spatially accurate information about soil fertility attributes, fertilizers and limes application may be optimized, increasing agricultural productivity, and reducing input costs. Nevertheless, it is important to recognize that there are still several steps to be overcome to reach this stage of development. Further studies are needed to develop protocols that simplify, for example, the accurate and reproducible processing of samples, spectral extraction, choice of experimental conditions and multivariate calibration algorithm for optimized modeling based on class, variability and collection procedure of soil, expansion of the database to promote greater robustness of predictions, etc.

4. Conclusion

The present study evaluated the optimized instrumental condition for soil fertility attributes employing a pXRF spectrometer and soil samples. Exploring the different possibilities of results from a commercial pXRF routine, four data matrices were used in the multivariate modeling. Three of them with raw spectra and one with elemental concentration data. The use of raw spectra measured with 15 kV and 12.15 μ A without filters as experimental conditions, modeled with PLS allows the best performance for the simultaneous quantification of the 10 soil fertility attributes evaluated. By the established RPD ranges [50],

the SOC, CEC, SB, and Ca^{2+} models were considered very good for quantitative prediction while the K^+ and Mg^{2+} models were considered good for quantitative predictions. The P_M , pH, BSP, and $\text{H} + \text{Al}$ models presented fair predictions, which may be used for evaluation and correlation. These results are reliable for local applications in the context of the soil samples evaluated in this study. However, they demonstrate the pXRF application potential combined with multivariate calibration, which may be applied in different new contexts to extract quantitative and qualitative information on soil fertility.

Conversely, the Relative Improvement and p -values comparing the optimized condition with data-fusion models demonstrated that a single experimental condition employment is sufficient to produce accurate and reliable results in this study context. Optimally utilizing this single experimental condition results in shorter time intervals for measurements, a significantly smaller amount of data to manipulate, and less processing power required to manage the models.

Additionally, the performance of the modeling with pXRF was considered equivalent to the EDXRF benchtop equipment by the randomization test, revealing a potentially possibility for the *in situ* application of this method.

In conclusion, the present study provided subsidies regarding the pXRF experimental condition choice in soil fertility attributes modeling. These findings demonstrated that the methodology may be employed in local soil analysis to speed up results in terms of measurement time and processing power to turn agile the decision related to the use of fertilizers and other products for soil correction. Now that the optimized condition are established, further studies are necessary to effectively evaluate the performance of this approach under a wider range of soil variability, especially when *in situ* application will be considered.

CRedit authorship contribution statement

José Vinícius Ribeiro: Conceptualization, Methodology, Software, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration, Funding acquisition. **Felipe Rodrigues dos Santos:** Conceptualization, Methodology, Validation, Formal analysis, Writing – original draft, Writing – review & editing. **José Francirlei de Oliveira:** Validation, Formal analysis, Writing – review & editing. **Graziela M.C. Barbosa:** Resources, Writing – review & editing, Supervision, Project administration, Funding acquisition. **Fábio Luiz Melquiades:** Conceptualization, Methodology, Software, Validation, Formal analysis, Resources, Writing – original draft, Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.sab.2023.106835>.

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